



Skeletal mechanism generation for high-temperature oxidation of kerosene surrogates

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ABSTRACT

A skeletal mechanism with 106 species and 382 reactions is developed from a detailed kerosene combustion chemical kinetic mechanism that includes 209 species and 1673 reversible reactions for a tri-component surrogate mixtures, consisting of *n*-decane, *n*-propylcyclohexane, and *n*-propylbenzene. The directed relation graph (DRG) method for skeletal mechanism reduction is applied as the first step for reduction of mechanisms with large numbers of species. A revised DRG approach (Z.Y. Luo, T.F. Lu, et al., *Energy Fuels* 24 (2010) 6283–6293) is compared with the original one and it is shown to be more stable than the original DRG method. The simplified iterative screening and structure analysis method (ISSA) is used subsequently based on the reduced mechanism generated by the DRG method to remove redundant species and reactions simultaneously. A minimal skeletal mechanism from the detailed mechanism for kerosene combustion is thus obtained. It is found that the discrete local reaction rates rather than the time-averaged reaction rates should be adopted when using the ISSA method for mechanism reduction due to large temperature difference and different reaction pathways in combustion at different simulation conditions. Although reduced in size by a factor of 2 for species and a factor of 4 for reactions, the skeletal mechanism exhibits high accuracy for high-temperature applications in predicting global combustion parameters, such as ignition delay, detailed profiles of species concentrations, and laminar flame speed. Furthermore, numerical simulation results of different mixture compositions are also comparable with those based on the detailed mechanism, indicating that the major reaction pathways of each component are captured by the reduced mechanism and the hierarchical structure of the detailed mechanism is maintained.

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1. Introduction

It is generally recognized that realistic numerical simulation of combustion must necessarily require both realistic descriptions of the laws of fluid mechanics and chemical kinetics of fuels [1]. Therefore, in recent years there has been an increasing effort to incorporate more realistic reaction mechanisms in numerical simulations beyond the simplistic descriptions assuming either equilibrium chemistry or one-step overall reaction. However, detailed combustion mechanisms of fossil fuels usually consist of hundreds of species and thousands of reactions and the employment of large detailed mechanisms in large scale numerical simulations demands huge amount of computational resources. Although the available computational power is growing rapidly, the computational cost associated with numerical simulations of turbulent combustion using detailed chemical kinetic mechanism

at practical conditions remains challenging. Therefore, in order to make numerical simulations of combustion computationally affordable and comprehensively reliable, the development of computational approaches for rigorous reduction of detailed chemical kinetic mechanisms is essential.

Kerosene is a mixture of a large number of hydrocarbon compounds with varying chemical structures, and it is not practical to model the combustion processes of every involved fuel component [2]. For simplification, a surrogate containing two to six pure compounds is usually adopted to mimic jet-fuel behaviors [3–7]. Such multi-component surrogates can be adjusted to match multiple experimental targets such as ignition delay, flame propagation speed, and physical properties of real kerosene. However, detailed chemical kinetic mechanism of kerosene combustion is highly complex, even for surrogate mixtures with only a few components. Therefore, numerical investigations with detailed combustion chemistry are severely restricted to some zero- and one-dimensional applications in which the governing equations of fluid mechanics are greatly simplified [2–7]. Consequently, these

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limitations have forced scientists to develop mechanism reduction methods for substantially decreasing the numbers of chemical species and reactions without significant loss of accuracy of the detailed mechanisms.

Mechanism reduction has been extensively studied and a variety of methodologies have been developed as reviewed in Ref. [1]. Mechanism reduction methods can be classified into two classes [8]: the first is skeletal reduction, which removes unimportant species and reactions from the detailed mechanism, usually referred as “model simplification”; the other approach produces a number of global reaction steps, whose rates are computed on the basis of the elementary reaction rates. The latter approach, usually referred as “model reduction,” can effectively be applied after skeletal reduction.

Skeletal reduction is typically the first step in mechanism reduction. It can be achieved through species reduction methods or by removing unimportant reactions from the detailed mechanisms. Elimination of redundant species and corresponding reactions from a large reaction mechanism can reduce the simulation time significantly, since the computational effort of simulation typically scales quadratically with the number of species, while only linearly with the number of reactions [1]. Various methods for species reduction have been reviewed by Nagy and Turanyi [9]. Species elimination now can be achieved through the computational singular perturbation (CSP) method [8], detailed reduction [10], level of importance (LOI) [11], connectivity method (CM) [9,12], element flux analysis [13], directed relation graph (DRG) [14–16] and other DRG-based methods such as DRG with error propagation (DRGEP) [17] as well as path flux analysis method [18], etc. The DRG-based methods combined with sensitivity-based methods, e.g., the DRG-aided sensitivity analysis (DRGASA) [19], and DRGEP with sensitivity analysis (DRGEP-SA) [20] are also developed to achieve a minimal skeletal mechanism with larger computational cost. Some of these methods, e.g., element flux analysis, and DRG are also incorporated into CFD simulations through adaptive reduction or on-the-fly reduction [21–24].

The species elimination procedure results in different reduced mechanisms with different number of necessary species, depending on the error of simulation tolerance and the simulation conditions where the skeletal mechanism is applied. After removal of redundant species, the reduced mechanisms can be reduced further via the elimination of unimportant reactions, and this can lead to a minimal skeletal mechanism, while the error of simulation remains essentially the same [9]. Methods for elimination of reactions include sensitivity analysis [25,26], principal component analysis (PCA) [27], computational singular perturbation (CSP) [8,28–32], genetic algorithm [33], etc. When skeletal mechanism has been achieved, model reduction methods, such as those based on time scale analysis [28–32], lumping [34–36], or quasi-steady state approximation [37,38] can be performed. Therefore, integrated strategies which combine these reduction methods in an appropriate sequence are often adopted to obtain a minimal skeletal mechanism or global reaction mechanism [1].

In the present work, we have used the DRG method combined with the iterative screening and structure analysis method (ISSA) [39,40] to construct a skeletal mechanism for combustion of kerosene on a large combustion domain by the removal of potentially redundant species and reactions without significant loss of the accuracy of the detailed mechanism. The different definitions of the direct interaction coefficient in DRG and the use of the ISSA method in combustion mechanism reduction are investigated. The paper is organized as following: in Section 2 we present the methodologies used for the mechanism reduction; results of the adopted methods and the performance of the skeletal mechanism are validated through a series of simulations including auto-ignition, perfectly stirred reactor (PSR), and 1-D flame propagation

simulation by comparing with the detailed mechanism in Section 3, and the main conclusions are briefly summarized in Section 4.

2. Methodologies

The kinetic model for combustion of kerosene used here is proposed by Dagaut et al. It consists of 209 species and 1673 reversible reactions [2]. The molar composition of surrogates used in this modeling is *n*-decane (NC10H₂₂) (74%), *n*-propylbenzene (PHC3H₇) (15%), and *n*-propylcyclohexane (CYC9H₁₈) (11%). The mechanism has been validated against a wide range of experimental results, especially the PSR experiments. Luche et al. [41] have used three different methods, i.e., element flux analysis, PCA and quasi-steady state approximation (QSSA) to reduce a three-component kerosene combustion mechanism. However, the reduced mechanism was generated and tested only based on PSR simulations. The skeletal mechanism derived in Luche's work by using element flux analysis and PCA methods involves 134 species and 1220 reactions. Further reduction to 33-species mechanism is obtained by introducing quasi-steady state approximation (QSSA). However, when a larger number of species are chosen to be in quasi-steady state, the validity of QSSA for some given species will be questionable in the entire simulation domain where temperature and pressure are beyond the pre-specified QSS conditions, and the traditional method for calculating the concentration of QSS species will be less efficient [1]. In the present work, an integrated method that combines DRG and ISSA methods will be employed to derive a reduced skeletal mechanism for combustion of kerosene, which can reduce the computational cost in numerical simulations significantly.

The DRG and ISSA methods are applied to reaction state points densely sampled from constant pressure auto-ignition simulations over a wide range of simulation conditions [42,43], and the important species and related reactions are captured by these two methods at each sample point. It has been previously confirmed that although reaction states are only sampled from auto-ignition in mechanism reduction, the generated skeletal mechanism still exhibited good performance in predicting the extinction turning point in PSR and laminar flame speed [11,20]. Though PSR and 1-D laminar flame simulations can be included in the data sampling for mechanism reduction and error control, the ignition demonstrates more restrictions for reaction eliminations than PSR [34]. In addition, auto-ignition simulations are much faster than 1-D laminar flame simulations. Consequently, in the present work, data sampling for mechanism reduction is only from auto-ignition simulation, and ignition delay time is selected as the only control parameter for error control in mechanism reduction [11,20]. The final reduced mechanism is the union of the sets of selected species and reactions. Another common feature in mechanism reduction is that the size of the reduced mechanism is controlled by a threshold value. Success of a mechanism reduction method can be judged by the size (number of species and reactions) of the reduced mechanism, the decrease in simulation time, and the error tolerance of the reduction. The latter is estimated based on comparison of the simulation results at sampling points using the full and the reduced mechanisms.

To achieve a skeletal mechanism suitable for jet-engine and other applications, the reduction is performed within the parameter range of pressure from 1 to 20 atm and equivalence ratio from 0.5 to 1.5. The initial temperature for constant pressure auto-ignition is set to be 1000–1600 K.

The reduced mechanism is tested over a wide parameter range in 0- and 1-dimensional applications that are relevant to high-temperature engine combustion. Based on the nature of the reduction approach which can maintain the major reaction path ways of each

surrogate component, the resulting reduced mechanism is also reliable for different mixture compositions, which will be validated through extended simulations.

2.1. Skeletal mechanism generation based on DRG

The automatic mechanism reduction procedure based on the DRG method recently proposed by Lu and Law [14–16] is an efficient method for skeletal mechanism reduction. It can be carried out with a linear scaling algorithm, in which the computational effort for mechanism reduction scales linearly with the number of species and reactions. Therefore, DRG is particularly suitable to be applied as the first step to reduce large kinetic mechanisms. More time-consuming methods can be subsequently applied to further reduce the skeletal mechanism obtained by DRG. In DRG method, the coupling pattern between species are represented by a directed graph, in which each vertex in the graph denotes a species in the detailed mechanism, and an edge exists from vertex A to vertex B if and only if the removal of species B would directly induce significant error to the production rate of species A . This effect is measured by the interaction coefficient, r_{AB}^{DRG} , originally defined as [14]

$$r_{AB}^{DRG} = \frac{\sum_{i=1,I} |v_{A,i} \omega_i \delta_B^i|}{\sum_{i=1,I} |v_{A,i} \omega_i|} \quad (1)$$

$$\delta_B^i = \begin{cases} 1, & \text{if the } i\text{th elementary reaction involves species } B \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$\omega_i = \omega_{f,i} - \omega_{b,i} \quad (3)$$

where r_{AB}^{DRG} is approximately the relative error induced to species A by elimination of species B , subscript i indicates the i th elementary reaction, $v_{A,i}$ is the net stoichiometric coefficient of species A in the i th reaction, and ω_i , $\omega_{f,i}$, and $\omega_{b,i}$ are the net, forward and reverse reaction rates, respectively. Starting from some important species, a species is identified as important if it is connected by an edge from an important species. This procedure is iterated until no more important species are chosen and the reduced mechanism is thus generated by including only important species and reactions involving them. The size of the reduced mechanism is controlled by a threshold value which decides the existence of the vertex. Recently, a revised DRG approach with a different definition for the interaction coefficient is proposed by Luo et al. [44] in order to handle the large isomer groups in biodiesel mechanism reductions. In this revised DRG method, the interaction coefficient is defined as [44]:

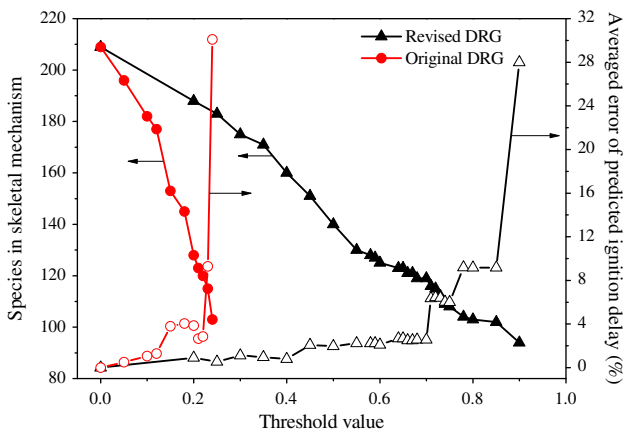


Fig. 1. Average absolute error of ignition delay time predicted using the reduced mechanism, and the number of species in skeletal mechanisms as a function of the threshold value for kerosene at 1 atm with equivalence ratio of 1.0.

$$r_{AB}^{DRG-new} = \frac{\max_i |v_{A,i} \omega_i \delta_B^i|}{\max_i |v_{A,i} \omega_i|} \quad (4)$$

In the present work, the difference of these two DRG methods on mechanism reduction is investigated. In the course of mechanism reduction, the error of the reduced mechanism is calculated based on the ignition delay time as defined in the following equation:

$$error_{rel} = \frac{|\tau_{ign,skel} - \tau_{ign,det}|}{\tau_{ign,det}} \times 100\% \quad (5)$$

where $\tau_{ign,skel}$ and $\tau_{ign,det}$ represent auto-ignition delay time due to the skeletal and detailed mechanism, respectively.

2.2. Iterative screening and structure analysis method

In the ISSA method, redundant species and redundant reactions are removed simultaneously [39,40]. Here, we have adopted a slightly modified ISSA method (without structure analysis) for mechanism reduction following Ref. [40]. In the ISSA method, the production and destruction rate of a species A is defined separately as

$$P_A = \sum_{i=1,I} \max(v_{A,i} \omega_i, 0) \quad (6)$$

$$D_A = \sum_{i=1,I} \max(-v_{A,i} \omega_i, 0) \quad (7)$$

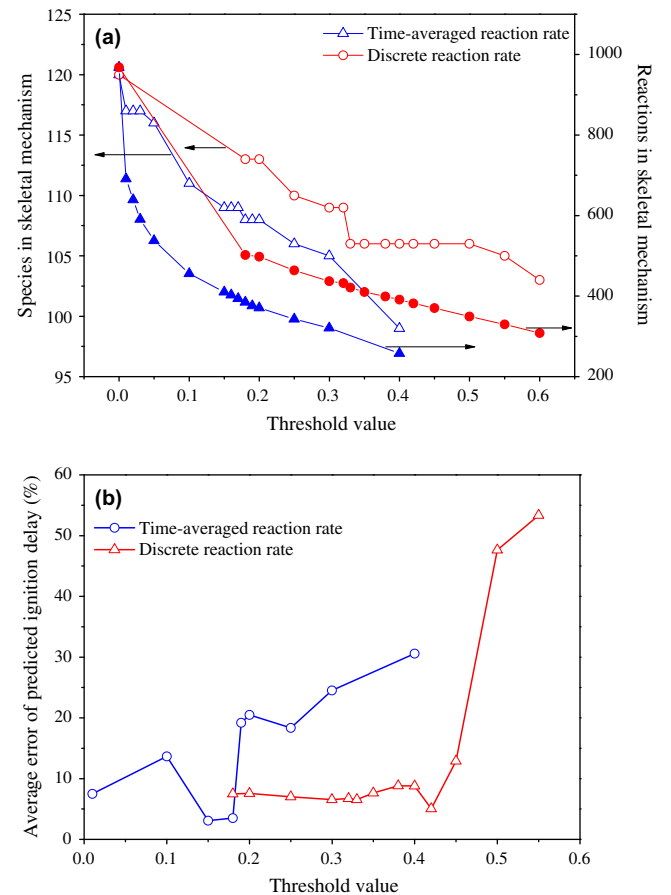


Fig. 2. The ISSA reduction results: (a) The number of species and reaction in skeletal mechanism at different threshold values using the time-averaged reaction rates and the discrete reaction rates, respectively; (b) Predicted average error of ignition delay time versus the threshold value for stoichiometric kerosene surrogates/air mixture from 1000 to 1800 K at 1 atm.

In order to estimate the relative importance of the i th reaction to species A , two normalized interaction coefficients are defined in ISSA as follows [39,40]:

$$f_{A,i} = \frac{\max(v_{A,i}\omega_i, 0)}{\sum_{i=1,I} \max(v_{A,i}\omega_i, 0)} = \frac{\max(v_{A,i}\omega_i, 0)}{P_A} \quad (8)$$

$$g_{A,i} = \frac{\max(-v_{A,i}\omega_i, 0)}{\sum_{i=1,I} \max(-v_{A,i}\omega_i, 0)} = \frac{\max(-v_{A,i}\omega_i, 0)}{D_A} \quad (9)$$

where $f_{A,i}$ is to evaluate the relative importance of the i th reaction as a production reaction to the overall production rate for species A , while $g_{A,i}$ represents the relative importance of the i th reaction as a destruction reaction to the overall destruction rate for species A . Important reactions and species are determined in an iterative procedure using a simplified version of the original methodology of Mauersberger [39] as follows:

First, the two interaction coefficients $f_{A,i}$ and $g_{A,i}$ for the initially selected target species are calculated; secondly, both $f_{A,i}$ and $g_{A,i}$ are sorted in a descending order, namely, the i th reaction with the largest $f_{A,i}$ or $g_{A,i}$ value is the first in the list, and the first M reactions are selected as important reactions if $\sum_{i=1,M} f_{A,i} \leq (1 - \varepsilon)$ and $\sum_{i=1,M} g_{A,i} \leq (1 - \varepsilon)$, where ε is a threshold ($0 \leq \varepsilon \leq 1$). Finally, both reactants and products in the selected important reactions identified from previous step are reformulated as the new group of important species. The above iterative procedure is repeated until no more species and reactions are added into the list of important species and reactions. Those species and reactions that are not selected are eliminated from the original mechanism.

The interaction coefficients defined in the ISSA method is similar to those in the recently proposed path flux analysis (PFA) method [18], both of which separate the net production rate of species into production and destruction terms. By using the production and destruction fluxes rather than the absolute reaction rate, Sun et al. [18] found that it is more accurate to capture the reaction path flux. It is found that the ISSA method is also efficient in capturing the major reaction pathways in mechanism reduction compared with the PFA method. The main difference is that in the PFA method contributions of other species to the production and destruction rates of the target species are dealt with, while in the ISSA method the contribution of reactions to the production and destruction rates of the target species is adopted to search important species and reactions simultaneously. ISSA may thus be able to provide skeletal mechanisms with minimal number of reactions and species while still maintains the major reaction pathways of important intermediates at the same time.

In the present work, we have implemented the original and the revised DRG method as well as the ISSA method based on the Chemkin 2.0 program package [42,43] and all the simulations are also performed with Chemkin 2.0.

3. Results and discussions

3.1. Results of DRG reduction

The DRG method is one of the most efficient reduction methods and is usually used as the first step in mechanism reduction. However, the skeletal mechanism generated by DRG is generally not

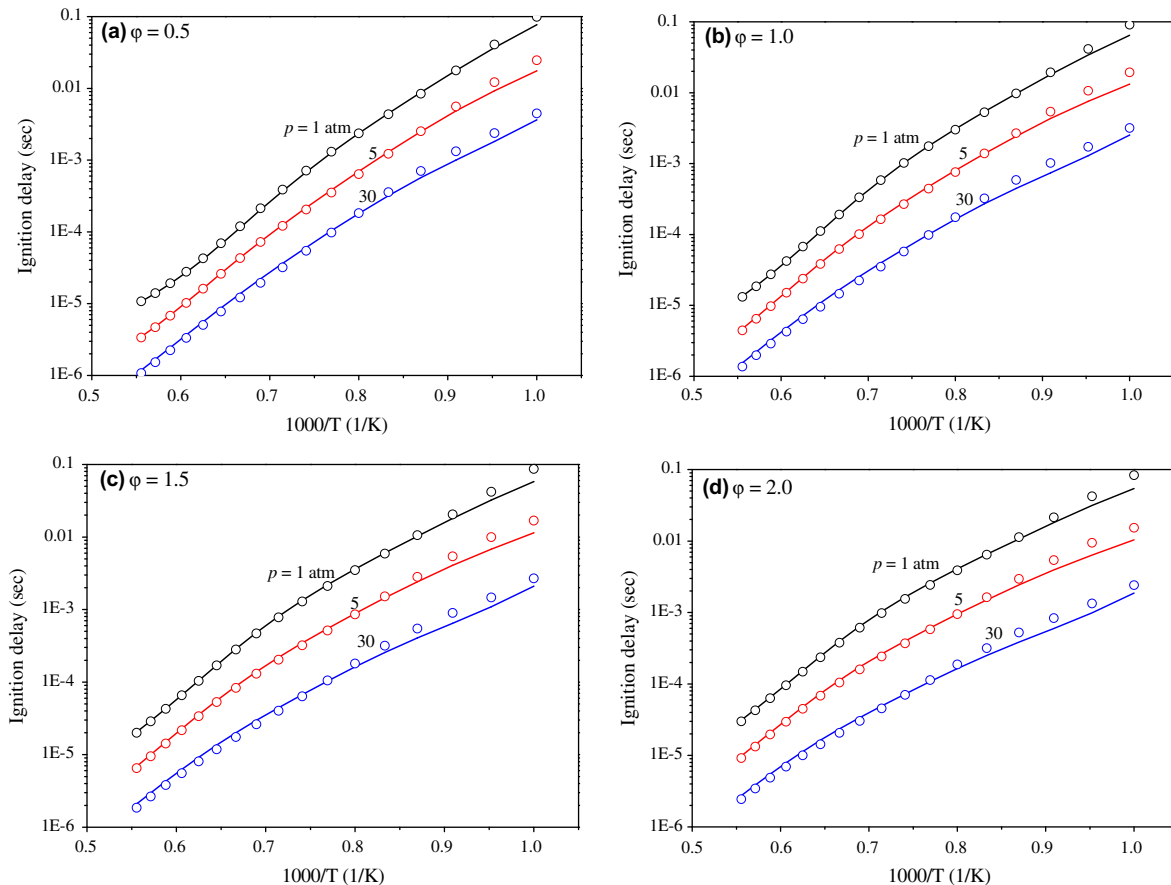


Fig. 3. Ignition delay time of the surrogate mixture for kerosene, as a function of the initial temperature for constant pressure auto-ignition, with the detailed (solid lines) and 106-species skeletal mechanism (symbols), respectively. The model-fuel molar composition used in simulation is *n*-decane (74%), *n*-propylbenzene (15%), and *n*-propylcyclohexane (11%).

minimal since it is assumed in the DRG method that all maintained species are of equal importance. Therefore, integrated strategy that apply other methods such as DRGASA and PCA methods in a successive sequences is often adopted to further reduce the skeletal mechanism generated based on the DRG method for maximal reduction. In the present work, we adopt the ISSA method to further remove redundant reactions and species simultaneously. The initially selected important species in the DRG method are *n*-decane, *n*-propylbenzene, *n*-propylhexane, O₂ and N₂, i.e., the reactants of combustion of kerosene in the detailed mechanism. The two definitions of the interaction coefficient in DRG shown in Eqs. (1) and (4) are both used to study the effect of different formulas on the mechanism reduction results. The number of species and the average error of the predicted ignition delay time with respect to the threshold value are shown in Fig. 1. One can see from Fig. 1 that as the threshold value increases, the average error of predicted ignition delay time increases. With the original DRG method, the average error of the predicted ignition delay time of the skeletal mechanism shows jump increases as the threshold value increases to larger than 0.22. This has also been found in other applications when the threshold value approaches 0.2 [14,18]. On the other hand, with the revised DRG method, neither the species number in the skeletal mechanism nor the error of predicted ignition delay time shows sudden jump until the threshold value reaches 0.7. And the error of predicted ignition delay tends to be more stable. Therefore, the revised DRG method proposed to treat large isomer groups in mechanism reduction demonstrates significant improvements in general cases. In addition, the results of the revised DRG method show similar tendencies as those of the PFA method [18]. The reason can be attributed as the *max* operator is used instead of the *sum* operator in the original work, therefore, the essence of the new definition is through the selection of the important reactions which show largest contributions to the production of major species to select species. Previously, similar normalization equation has been used in the time-integrated element flux analysis [45], and it is shown that this definition is very helpful to report all possible transformations relative to the fuel decomposition channel. Therefore, even when large threshold value and error tolerance are used, this definition still performs well. And the definition by Eq. (4) in DRG should be particularly useful in detailed mechanism reduction for pyrolysis of fuels. This can be explained by an illustrative example. For mechanism reduction of *n*-decane, we found that even large threshold value (e.g., 0.75) is used, the major intermediates of the fuel decomposition channels, five decyl radicals can be captured at the initial state during ignition simulation via the H-abstraction reactions which dominate the consumption of *n*-decane. To sum up, it can be expected that in DRG and related methods, different formulations for the interaction coefficient from the original definition, such as separating the contributions of the reactions or species to the production and consumption of major species (PFA method [18]), can improve the performance of the DRG method. The performance of these different definitions for the direct interaction coefficient on their ability to the treatment of partial equations, fast processes, and quasi-steady state species merits further study.

Though the revised DRG method performs better than the original one, it can be seen that at the same error tolerance of 5% or smaller, the original DRG method is still very competitive compared with the revised one. And the original definition in DRG can also handles the fast reactions and quasi-steady state species well in skeletal mechanism reduction [16]. Therefore, the skeletal mechanism obtained from the original DRG method with a threshold value of 0.22 containing 120 species and 968 elementary reactions is used for further reduction. We have found that few species can be removed even with a large threshold value (0.4) if a second DRG reduction is performed based on the reduced mechanism. This

shows that there exist strong correlations between species in the reduced mechanism. The average error of ignition delay for stoichiometric kerosene surrogates/air mixture from 1000 to 1800 K at 1 atm predicted by the skeletal mechanism is 2.86%, while the maximum relative error is 18% at 1000 K. This indicates that the relative error is mainly caused by the low-temperature regions. It should be noted that the original detailed mechanism is developed only for high-temperature applications and the skeletal mechanism should thus not be used when the temperature is below 1000 K, e.g., the negative temperature coefficient (NTC) regions.

3.2. The results of the ISSA method and discussions

The ISSA method is used to generate the minimum skeletal mechanism based on the reduced mechanism given by the DRG method. The ISSA method was previously applied to reduce mechanism of chemical kinetic processes involved in the atmosphere environment [39,40]. In the original ISSA method, Mauersberger [39] proposed to use time-averaged reaction rates instead of the reaction rates on discrete time to produce a smaller reduced mechanism which is validated by latter studies [40]. However, in the present work we find that discrete local reaction rates rather than time-averaged reaction rates should be used. This may be due to the large temperature range and different reaction pathways in combustion at different simulation conditions. Figure 2 shows the number of species and reactions in the skeletal mechanisms as well as the average error versus the threshold value. It can be seen that with the increase of the threshold value, the number of

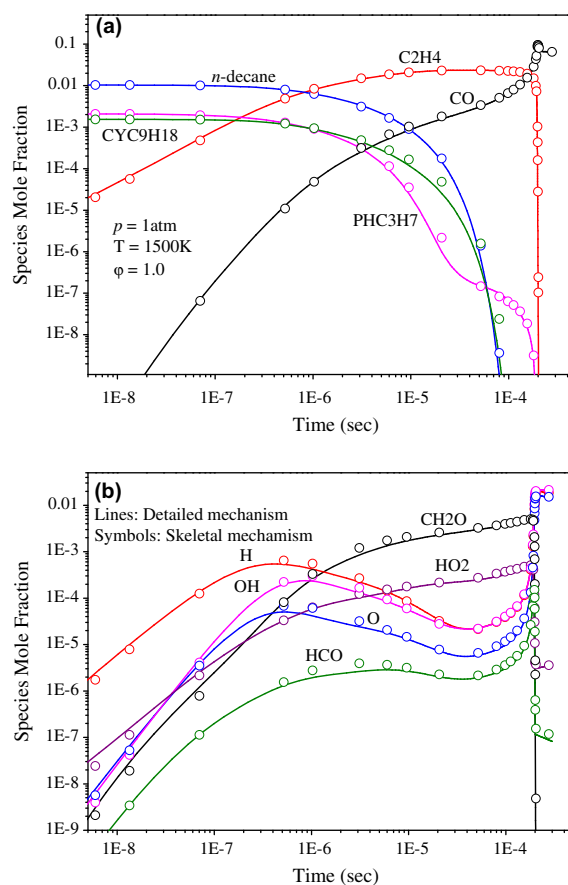


Fig. 4. Species profiles including major fuel species and minor radicals, in a constant pressure auto-ignition simulation, calculated with the detailed (solid lines) and 106-species skeletal mechanisms (symbols), respectively.

species and reactions in the skeletal mechanism decreases as expected. However, when using the time-averaged reaction rates, the average error increases from less than 5% to more than 20% when the threshold value is approaching 0.2. On the other hand, the average error tends to be more stable until the threshold value reaches 0.45 using discrete local reaction rates. It is also demonstrated in Fig. 2 that at the same threshold value, although the number of species and reactions in the obtained skeletal mechanism using time-averaged reaction rates is smaller than that with discrete reaction rates, the error of the predicted ignition delay time is larger. This means the number of species and reactions in the skeletal mechanism generated by the ISSA method using time-averaged reaction rates is actually larger than that with discrete reaction rates in order to achieve the same error tolerance for the predicted ignition delay time. Furthermore, the errors of the ISSA with time-averaged reaction rates do not increase monotonously with the increasing threshold value. Therefore, unreliable skeletal mechanisms may be generated with the ISSA method using time-averaged reaction rates, which also indicates that discrete local reaction rates should be used in order to capture different reaction pathways under various reaction conditions. By using the discrete reaction rates in the ISSA method with a threshold value of 0.42, 14 species and 586 reactions are removed, and the final skeletal mechanism containing 106 species and 382 elementary reactions is generated. It should be mentioned that the ISSA method can also be used alone for mechanism reduction, since it can eliminate species and reactions simultaneously while still maintaining major reaction pathways under different simulation conditions. Therefore, it will be valuable especially for adaptive

or on-the-fly reduction in CFD simulations. In the present work, the main objective is to obtain a validated skeletal mechanism for kerosene that can be used over a wide range of simulation conditions and the DRG method combined with the ISSA method is adopted for maximal reduction.

3.3. Reaction path analysis

The ISSA method provides an efficient way for mechanism reduction while still maintains major reaction pathways and corresponding reactions at different conditions. In order to illustrate this, we analyze the detailed ISSA results at 1400 K, 1 atm with equivalence ratio of 1.0 during auto-ignition simulation. The initially selected target species are *n*-decane, *n*-propylbenzene, *n*-propylhexane, O₂ and N₂. The first ISSA analysis is performed at time 3.0×10^{-6} s for these species, and the main reactions for the four species (N₂ is the dilute gas) are selected. For example, the important reactions for *n*-decane (represented by NC10H22) and *n*-propylbenzene (PHC3H7) according to the ISSA analysis are:

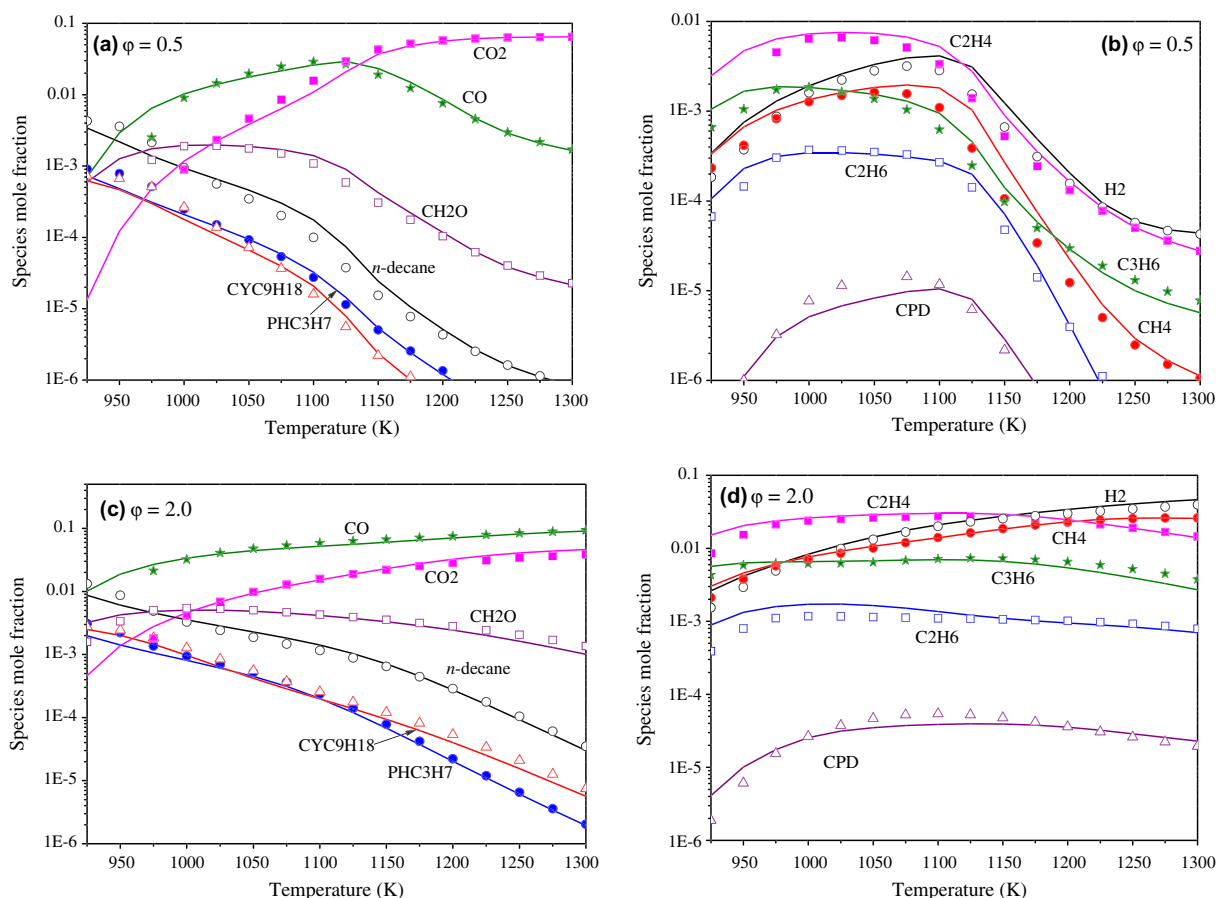
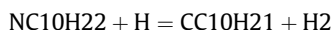
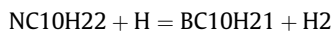
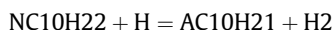


Fig. 5. Species profiles in PSR as a function of temperature for the kerosene surrogate mixture, calculated with the detailed (solid lines) and 106-species skeletal mechanism (symbols), respectively, with equivalence ratio of 0.5 and 2.0, at a pressure of 1 atm, and residence time of 0.07 s under fixed temperatures.

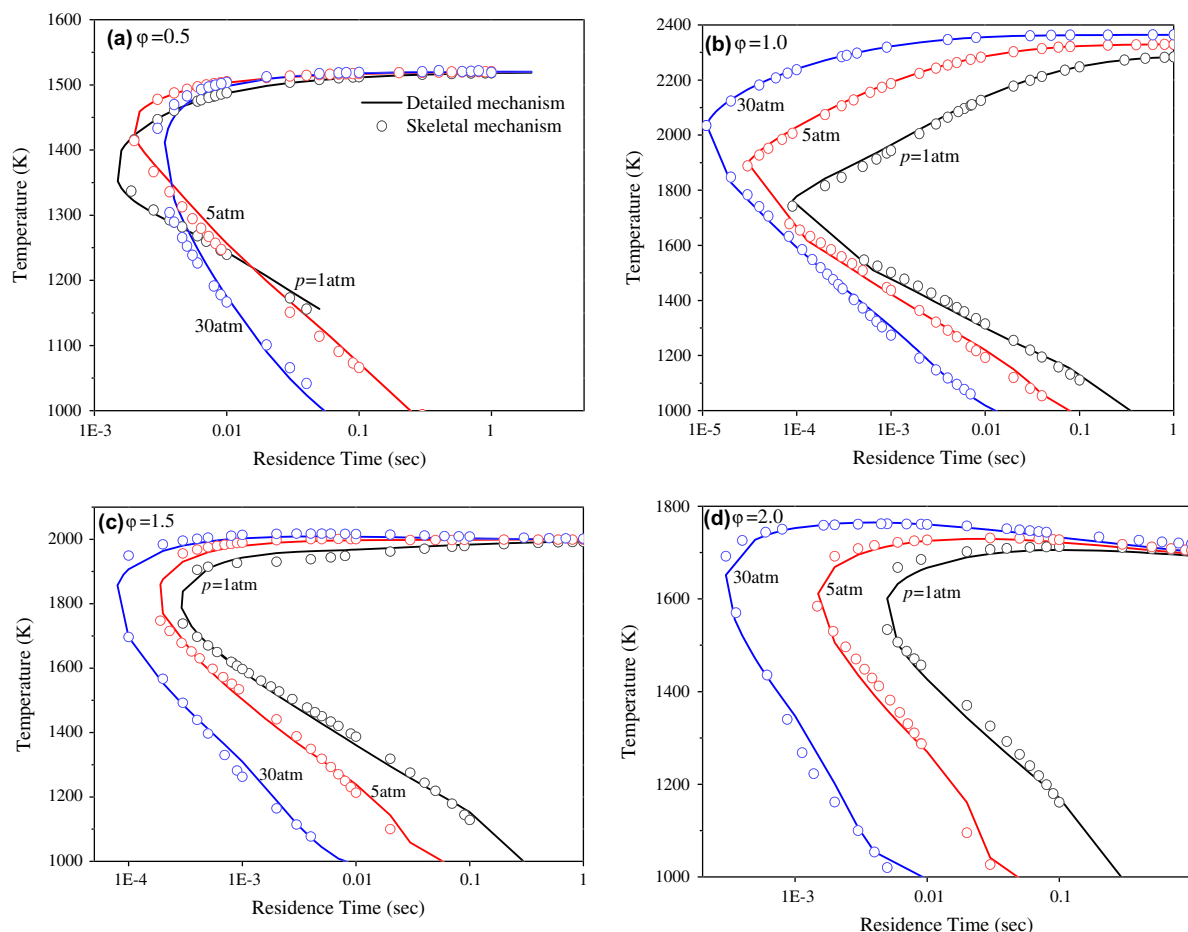
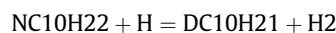


Fig. 6. The temperature profiles of PSR at different residence time for the kerosene surrogate mixture with various equivalence ratios and pressures at fixed inlet temperature of 300 K, calculated with the detailed (solid lines) and 106-species skeletal mechanism (symbols), respectively.



Therefore, the major intermediates, 1-octyl (AC8H17), 1-hexyl (AC6H13), 1-butyl (PC4H9), 1-decyl (AC10H21), 2-decyl (BC10H21), 3-decyl (CC10H21), 4-decyl (DC10H21), together with the abundant small radicals and molecules, C_2H_5 and H_2 , as well as the dominant reactions involving the direct decomposition of *n*-decane and H-abstraction reactions are selected. 5-decyl (EC10H21) is also captured through further iterative analysis via the isomerization reaction of 1-decyl (AC10H21). According to a previous study by Sun, etc. in an analysis of the PFA method [18], these species, e.g., AC10H21, BC10H21, CC10H21, and DC10H21 are very important in reproducing the reaction path fluxes. According to our analysis, they are also captured in the ISSA analysis directly through identifying the important H-abstraction reactions. For *n*-propylbenzene, the reaction of direct decomposition to produce benzyl radical (PHCH2) and C_2H_5 is found to be crucial due to the ISSA analysis, and this was also confirmed by previous sensitivity analysis [46]. These species will be considered as the new target species and the ISSA analysis will be performed for these molecules subsequently. Through an iterative procedure, the species and reactions which dominate the major reaction pathways at this condition are thus chosen. Consequently, the skeletal mechanism obtained from the ISSA method will provide reasonable results for the chemical kinetics of the major intermediates. For example, toluene is an important species during the oxidation

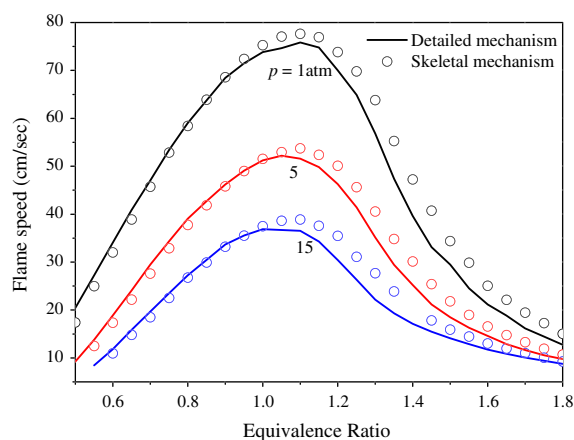


Fig. 7. Laminar flame speed versus equivalence ratio predicted by the detailed and skeletal mechanism at fresh gas temperature 473 K for pressure $p = 1, 5, 15$ atm, respectively.

of *n*-propylbenzene and can be identified through further iterative screening analysis of the newly selected species PHCH2. When using toluene as a new target species, the major reactions captured by the ISSA method indicated in the following were also found to be important by sensitivity analysis in the PSR simulations conditions [47]:

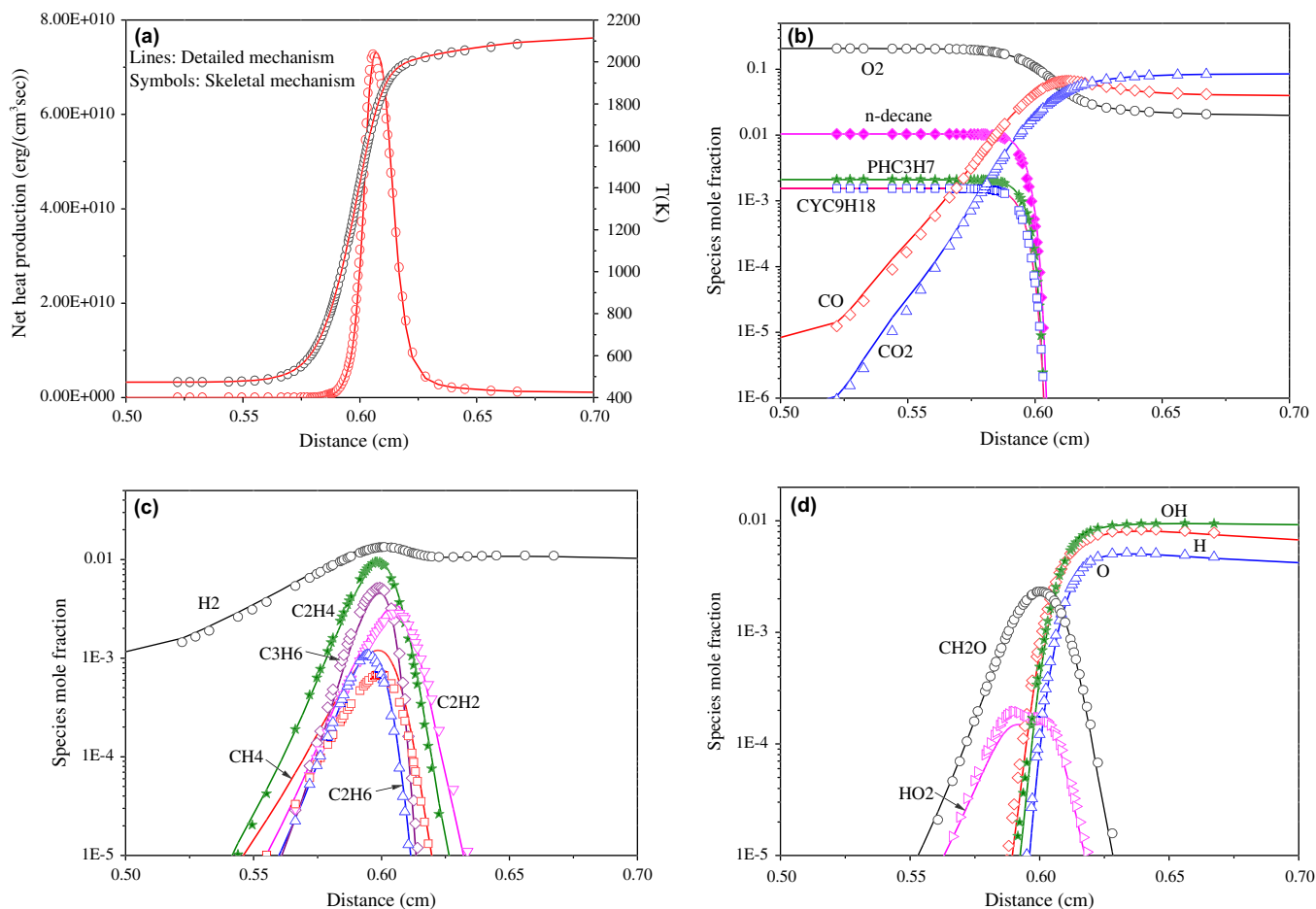
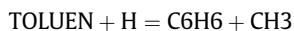
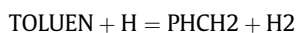
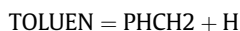
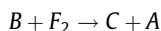
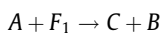


Fig. 8. Spatial profiles in a 1-D planar freely propagating laminar premixed flame, calculated with the detailed and the 108-species reduced mechanisms, for net heat production, temperature, major species, selected intermediate species, and radicals at a stoichiometric mixture under atmospheric pressure and initial inlet temperature of 473 K.



This means the skeletal mechanism generated by ISSA based on the mechanism of kerosene combustion can even be used for the oxidation of toluene at conditions where the reductions are performed. Thus, through this iterative procedure, the hierarchical structure of the detailed mechanism is maintained in the reduced mechanism. Similar to the PFA method, the ISSA method can also be useful for mechanism reduction involving catalytic species. For example, supposing that A and B are two catalytic species which have a relatively small net reaction flux, the following two reactions show both large comparable reactions rates in the oxidation of F_1 and F_2 . The catalytic effect which is important would not be captured by the DRGEP method, since the changes of the concentrations of the catalytic species, A and B , are very small.



However, these two reactions can be identified easily in the ISSA method as important reactions due to their large contribution to the destruction to the initially selected important species F_1 and

F_2 . The two catalytic species will thus be recognized as the new target species. Through an iterative procedure based on the ISSA method, major reaction pathways related to the catalytic species can be identified after the two catalytic species is chosen as target species. A typical and simple example which has been analyzed based on the PFA method in Ref. [18] is the catalytic effect of NO/NO_2 in the ignition of hydrocarbon fuels. Following a similar approach, it can be easily shown that reactions with large sensitivity coefficient together with the involved catalytic species NO/NO_2 can also be captured easily by the ISSA method. The ISSA method is thus competitive for catalysis mechanism reduction compared with other methods such as the time-consuming sensitivity analysis combined with the PCA method [48,49].

3.4. Skeletal mechanism validations

The 106-species skeletal mechanism has been validated against the detailed mechanism in homogeneous applications including the auto-ignition, perfectly stirred reactor (PSR), and 1-D laminar flame speed simulations. Figure 3 shows the predicted ignition delay time of the mixture composed of the same mole fraction as the original compositions used for mechanism reductions. It is found that the skeletal mechanism predicts the ignition delay time accurately over a wide range of parameters covering pressures from 1 to 30 atm, temperatures above 1000 K, and equivalence ratios of 0.5–2.0. Similar to the previous work on high-temperature skeletal mechanism generation [11,18–20], slightly larger discrepancies

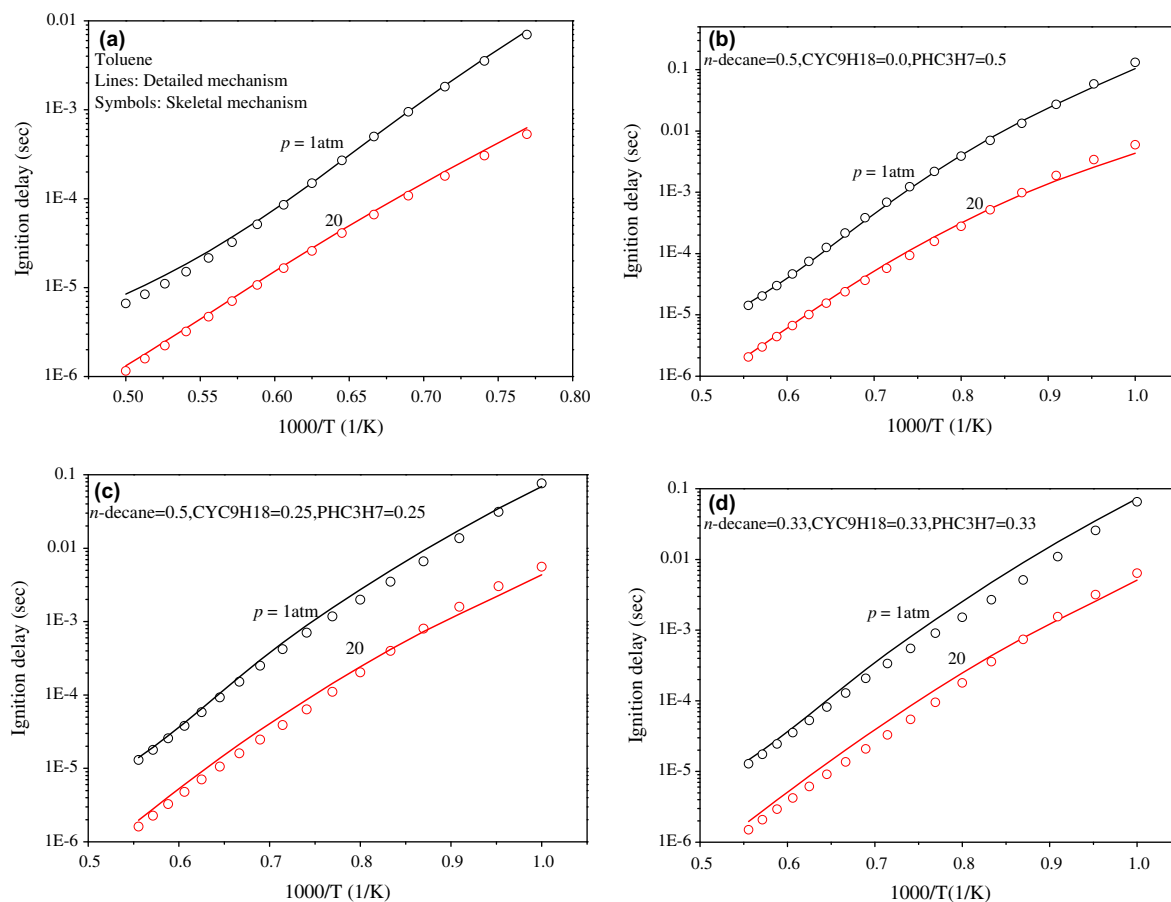


Fig. 9. Predicted ignition delay time of the surrogate mixture for kerosene, as a function of the initial temperature for constant pressure auto-ignition, with the detailed and 106-species skeletal mechanisms, respectively, with equivalence ratio of 1, pressure at 1 and 20 atm for different fuel compositions.

occur at low initial temperature ($T_0 = 1000$ K). This may be because most reaction state points based on which mechanism reduction is performed are taken from on high-temperature simulation results since the original mechanism is only suitable for modeling high temperature combustion of kerosene. It is well known that important reactions and species are different in high temperature and low temperature combustion. Some reactions or species which are important in vicinity of 1000 K may thus be eliminated with only a negligible effect on the averaged error of the ignition delay time. However, the skeletal mechanism still performs well within the temperature ranges from 1000 to 1800 K. Consequently, the reduced skeletal mechanism should not be used for temperatures below 1000 K, e.g., the negative temperature coefficient (NTC) regions. The major species profiles are also compared with those due to detailed mechanism as a function of residence time in auto-ignition simulations of a stoichiometric mixture under atmospheric pressure and initial temperature of 1500 K as shown in Fig. 4. The results obtained from the skeletal mechanism also agree well with those of the detailed mechanism.

To further demonstrate the performance of the skeletal mechanism, species profiles in perfectly stirred reactor (PSR) [50] are also compared between detailed and reduced mechanisms. Figure 5 shows the calculated mole fraction of some important species as a function of temperature in PSR using the detailed and skeletal mechanism, respectively, with equivalence ratio of 0.5 and 2.0 at a pressure of 1 atm, and residence time of 0.07 s. It can be seen that the mole fractions of important species resulting from the skeletal mechanism are rather close to those from the detailed mechanism, especially at high temperatures, while the error increases as the temperature decreases.

Extinction is another important property to validate chemical kinetic mechanisms and is also very important for kerosene combustion in jet engines. Therefore, temperature profiles as a function of the residence time in PSR over a wide range of equivalence ratios and pressures using the detailed and skeletal mechanisms are compared in Fig. 6. It can be seen that the skeletal mechanism shows good performance compared with the detailed mechanism on the combustion S-curves including both the branches above the turning points, which are known as the physically stable branches, and the branches below the turning points, known as the physically unstable middle branches. This indicates that the reduced mechanism can be also used for unsteady combustion simulations.

In order to validate the skeletal mechanism in flame simulation involving transport properties, the laminar flame speed together with important species profiles during 1-D flame simulations [51] are evaluated using the detailed and 106-species skeletal mechanism, respectively. Figure 7 shows the laminar flame speed as a function of the equivalence ratio covering pressures of 1, 5, and 15 atm with initial inlet temperature of 473 K. It can be seen that the skeletal mechanism also predicts the laminar flame speed accurately compared with the detailed mechanism except that larger discrepancies are found at the fuel-rich side. The net heat production and temperature changes during flame simulation together with the selected important species profiles are shown in Fig. 8. Again the results of the detailed and skeletal mechanism correlate well with each other, indicating that the reduced mechanism derived from spatially homogeneous systems using the present methods can typically be extended to diffusive systems.

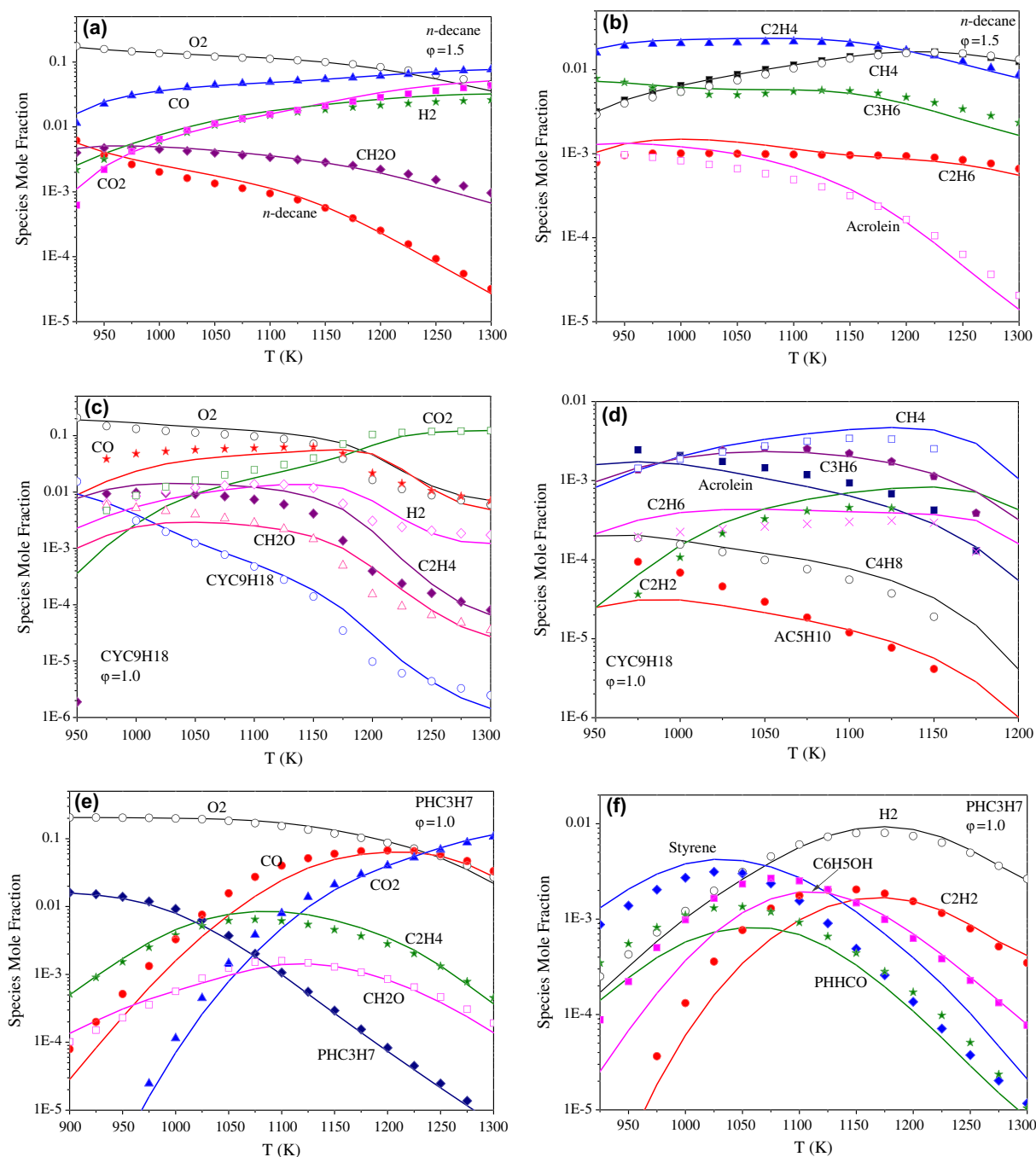


Fig. 10. Mole fraction of selected important species and intermediates as a function of temperature in the oxidation of each single component in the mixture used for kerosene surrogates in PSR simulation by using the skeletal and detailed mechanism at pressure 1 atm and residence time 0.07 s.

In the present work, we do not try to compare the results of the reduced mechanism with the experimental results since the purpose of this work is mainly to derive a skeletal mechanism that can be used under a wide range of simulation conditions. The skeletal mechanism is thus mainly validated against results of the detailed mechanism. Furthermore, the detailed mechanism has already been validated against experimental results for species profiles in PSR as a function of temperature with equivalence ratio of 0.5 and 2.0 at a pressure of 1 atm, and residence time of 0.07 s [2,3]. The species profiles in PSR under these conditions due to the skeletal mechanism agree very well with those based on the detailed mechanism for as shown in Fig. 5. As for experimental ignition delay time for kerosene, large discrepancies exist between different experiment results [2,52].

3.5. Extended mechanism validation

Since the present integrated strategy can capture major reaction pathways for each fuel component of the kerosene surrogates, it can be expected that the current skeletal mechanism can be used for different amount of surrogate components adjusted for mimicking real fuels, and the performance of the skeletal mechanism for different mixture compositions is validated. The constant pressure auto-ignition delay time for different fuel compositions as well as a major stable intermediate, toluene, during the oxidation of *n*-propylbenzene is shown in Fig. 9. It can be seen from Fig. 9 that the reduced mechanism reproduces closely the behavior of the detailed mechanism. The oxidation of the three single component used for constructing the kerosene surrogates in a perfectly

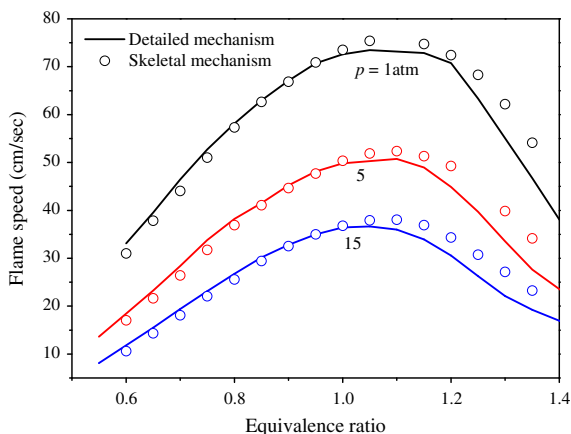


Fig. 11. Laminar flame speed for a two-component kerosene surrogates, *n*-decane(80%)/*n*-propylbenzene(20%) at initial inlet temperature of 473 K under different pressures.

stirred reactor is the next extended validation of the skeletal mechanism. The oxidation of *n*-decane, *n*-propylhexane, and *n*-propylbenzene has been extensively studied in PSR by experiment and modeling covering a wide range of parameters. Figure 10 illustrates some selected species profiles as a function of temperature in the oxidation of these single fuels in PSR with the detailed and skeletal mechanism. The predicted species profiles of major species and intermediates agree well compared with those due to the detailed mechanism. The final validation of the skeletal mechanism is the laminar flame speed of the two component surrogate, *n*-decane(80%)/*n*-propylbenzene(20%), because similar two-component model consisted of *n*-decane and some aromatic compounds (e.g., *n*-propylbenzene, 1,2,4-trimethylbenzene) is often selected as kerosene surrogates [2,5]. Figure 11 shows the predicted laminar flame speed versus equivalence ratio at initial gas temperature of 473 K. Good agreements are observed again, which demonstrates the present skeletal mechanism features good extensibility. These results indicate that the strong hierarchical structure embedded in the detailed mechanism is maintained in the skeletal mechanism.

4. Conclusion

The DRG combined with the ISSA method is successively applied to reduction of a detailed chemical kinetic mechanism for kerosene combustion. The original definition and the new definition of the direct interaction coefficient in DRG is compared and discussed. It is shown that the new definition can improve the performance of the DRG method, and the error of predicted ignition delay time tends to be more stable when more species are removed from the detailed mechanism. In previous ISSA method for mechanism reduction at atmospheric condition, time-averaged reaction rate over discrete local reaction rate was preferred. However, according to our analysis, the local discrete sampling reaction rate should be used when applying the ISSA method in combustion because reaction pathways differ significantly at different time or space. The ISSA method is similar to the recently proposed PFA method in that both methods can be useful in catalytic mechanism reduction. Both the PFA method and the ISSA method can capture major reaction pathways, which may be due to a separate treatment of contribution of reactions or other species to the production and destruction fluxes of the target species instead of using the absolute reaction rate to define the interaction coefficient between the species or reactions. However, the different definitions used for the evaluation of the dependence of species or reactions

to the major species in DRG and related methods still merit further study including their performance in dealing with fast processes, partial equations, and quasi-steady state species.

Based on an integrated strategy for mechanism reduction which combines the DRG and ISSA method, a 106-species and 382-reactions skeletal mechanism is developed from a detailed mechanism for kerosene surrogate. The skeletal mechanism is validated against the detailed mechanism over a wide range of parameters in homogeneous applications. Good agreements are obtained compared with the detailed mechanism for ignition delay, detailed profiles of species concentrations in PSR and laminar flame speed. The skeletal mechanism is also validated for various fuel compositions. The results obtained from the skeletal mechanism are rather close to those due to the detailed mechanism, which indicates that major reaction pathways are captured by the integrated reduction strategy used in the present work. Slightly larger discrepancies are found between the detailed and skeletal mechanism as the temperature decreases. However, it should be noted that the original detailed mechanism is developed for high-temperature applications, the skeletal mechanism is thus only suitable for high-temperature applications such as flame simulations.

Acknowledgments

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Appendix A. Supplementary material

The reduced skeletal mechanism is supplied as supporting information given as chemkin format.

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2011.05.019](https://doi.org/10.1016/j.combustflame.2011.05.019).

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